ASSESSMENT TOOL 2

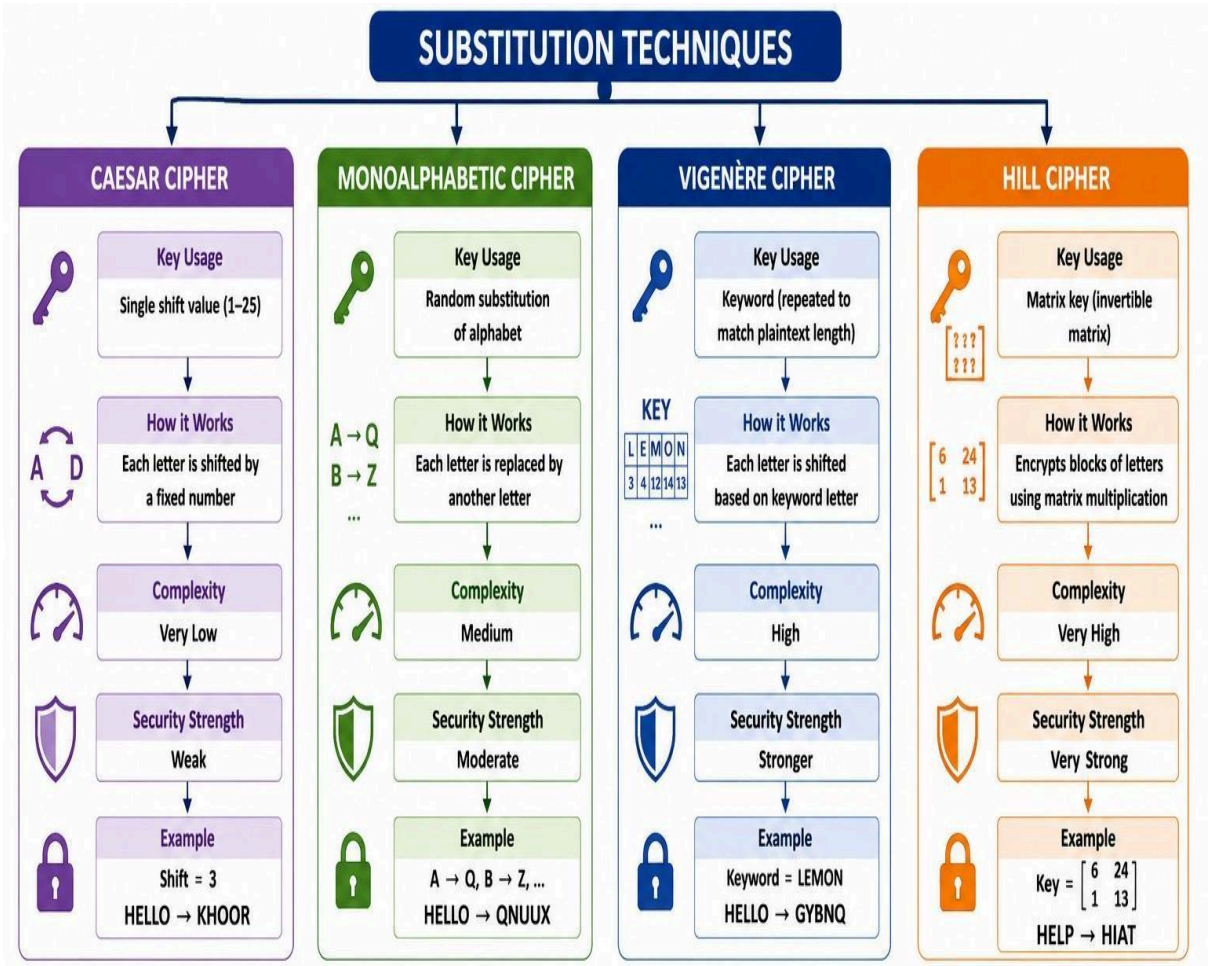
Concept Maps and Analysis



SECURE INFORMATION, SECURE FUTURE.

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Course Name	:	Cryptography and Network Security

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks: a. Show how each attack leads to specific threats and risks b. Map how security goals help mitigate them Provide one real-world example for each link.



COMPARISON TABLE

Cipher Technique	Key Usage	Complexity	Security Strength	Key Size / Space
Caesar Cipher	Single shift value	Very Low	Weak	25
Monoalphabetic Cipher	Random alphabet	Medium	Moderate	26!
Vigenère Cipher	Keyword	High	Stronger	26^k (k = key length)
Hill Cipher	Matrix key (n x n)	Very High	Very Strong	$(26^{\wedge}n^{\wedge}2)$ for n x n matrix

ANALYSIS – WHICH IS MOST SECURE & WHY?

Hill Cipher is the most secure among classical substitution techniques.

Reasons:

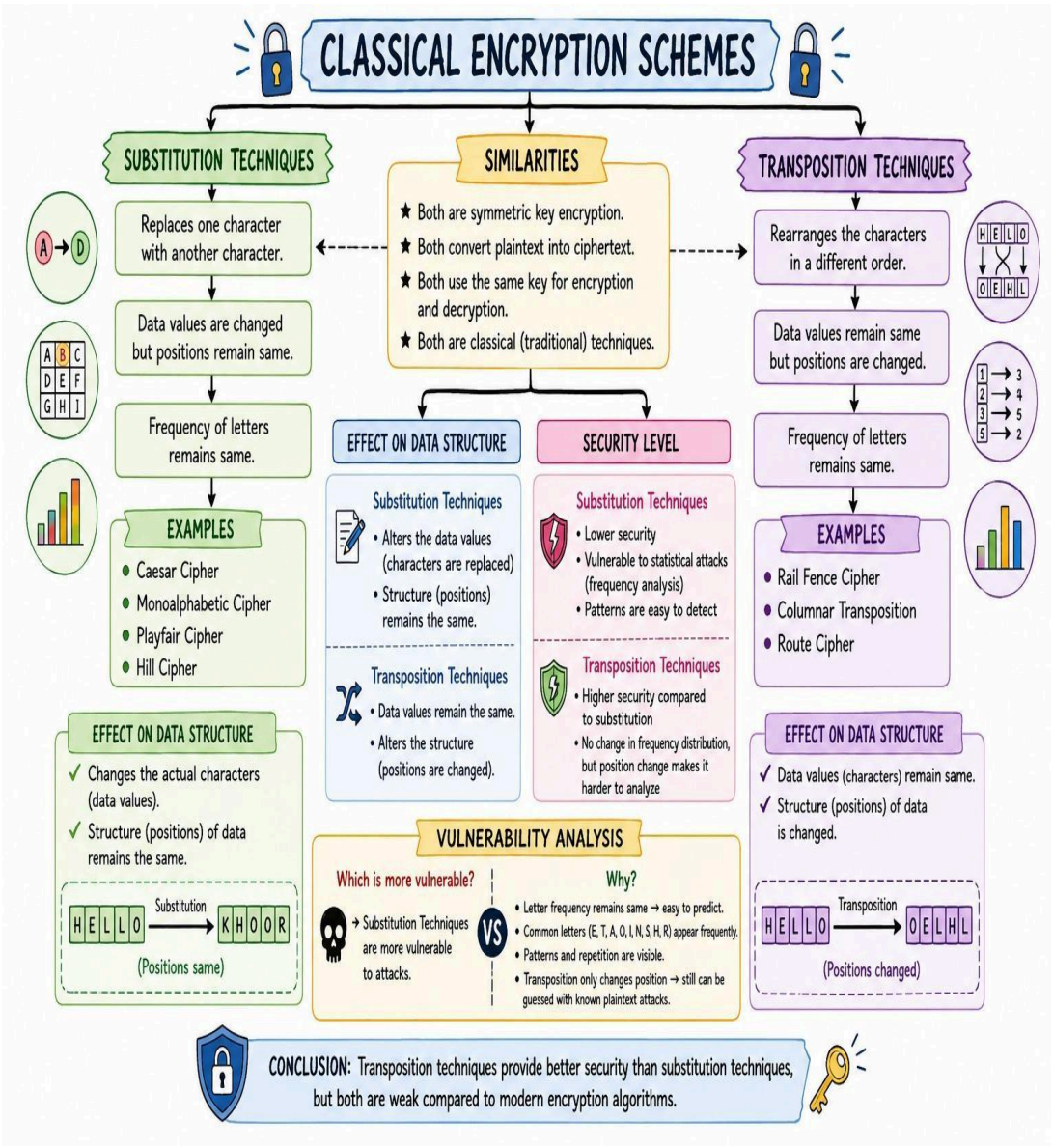
- Uses matrix multiplication on blocks of letters.
- Larger key space due to matrix size.
- Resists simple frequency analysis.
- More complex mathematical structure.
- Even slight change in key produces completely different ciphertext.

However, modern ciphers (e.g., AES) are far more secure than all these.

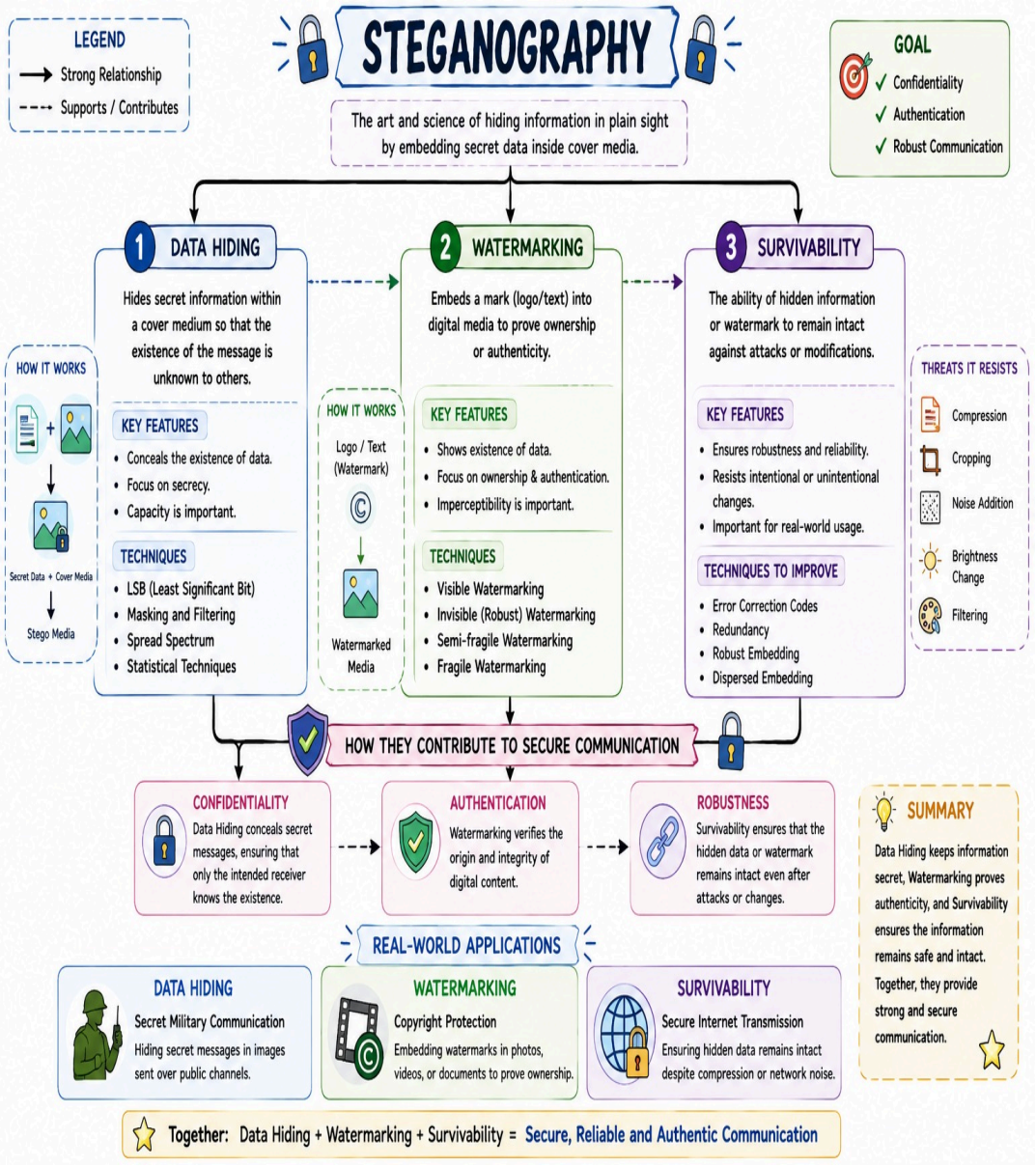
2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.



3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks: a. Show differences and similarities i. Map how each affects the Data structure and Security level ii. Analyze which is more vulnerable to attacks and why?



4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks: a. Show relationships between these concepts b. Analyze how each contributes to secure communication c. Include one real-world application for each.



5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks: a. Show how each concept is mathematically connected b. Map their role in encryption/decryption c. Analyze which concept is most critical for public key cryptography and justify.

